A whole lot of reductions

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1 Preliminaries

Definition 1. *A **tour** in a graph $G = (V, E)$ is a sequence of vertices and edges adjacent to them in the graph that spans every vertex. We usually refer to the multiset of edges defined by a tour as T. The **cost** of a tour T is the sum of the costs of the edges in T, denoted as $c(T) = \sum_{e \in T} c(e)$.*

This definition is equivalent to saying that a tour is a Eulerian multigraph that spans every vertex of G.

Definition 2 (Metric TSP)**.** *Let $G = (V, E)$ be a connected undirected graph. The **Metric Traveling Salesperson Problem** (Metric TSP) is defined as follows:*

- ***Instance:*** *A complete graph $G = (V, E)$ which is connected by definition, and a cost function $c : E \to \mathbb{R}_{\geq 0}$ that satisfies the triangle inequality:*

$$c(u, v) \leq c(u, w) + c(w, v) \quad \text{for all } u, v, w \in V. \tag{1}$$

- ***Objective:*** *Find a closed tour C in G such that the total cost $c(C) = \sum_{e \in C} c(e)$ is minimized.*

Definition 3 (Quasi-Tour)**.** *A **quasi-tour** T_0 in a graph $G = (V, E)$ is a subset of edges in T removing edges from a finite set of closed tours. Each connected component of this multigraph is either a Eulerian multigraph or an isolated vertex.*

A given multiset of edges T_0 is a quasi-tour if and only if it is Eulerian. In the rest of this article, we are going to use a slightly modified version of this problem, enforcing some edges.

Definition 4 (TSP_f)**.** *An instance of the **TSP_f** problem is defined through a weighted metric graph $G = (V, E_u \dot\cup E_f, c)$ Its objective is to find a tour that contains all the edges in E_f at least once and uses each edge at most twice.*

We say that a tour T satisfying the conditions of TSP_f is a **valid tour**.

Lemma 1. *Let $\varepsilon > 0$. There is a polynomial time reduction mapping a given instance (G, c) of TSP_f to an instance (G', c') of Metric TSP such that, if T is the minimum cost spanning tour in G and T' is the minimum cost spanning tour in G', then*

$$c(T) - \varepsilon \max_{e \in E_f} c(e) \leq c'(T') \leq c(T)$$

Then, we are not worsening the approximation factor of TSP by using TSP_f instead of TSP, as the optimal solution of TSP_f is at most $\varepsilon \max\limits_{e \in E_f} c(e)$ away from the optimal solution of TSP. For low enough values of ε, this difference is negligible, and thus the approximation factor is preserved.

2 Starting point

For the reduction, we will start from a common hard to approximate problem:

Definition 5 (MAX-E3-LIN-2)**.** *The **MAX-E3-LIN-2** problem is defined as follows:*

- ***Instance:*** *A set of n boolean variables $x_1, x_2, \dots, x_n$ and a collection of m linear equations over $\mathbb{F}_2$, where each equation involves exactly three variables. Each equation is of the form:*

$$x_{i_1} + x_{i_2} + x_{i_3} = b_j, \tag{2}$$

  where $b_j \in \{0, 1\}$ and i_1, i_2, i_3 are distinct indices from $\{1, 2, \dots, n\}$.

- ***Objective:*** *Find an assignment to the variables that maximizes the number of satisfied equations.*

Theorem 1. *For every $\varepsilon \in (0, \frac{1}{4})$ and a sufficiently large integer $k \geq k(\varepsilon)$, the following partial decision subproblem $Q(\varepsilon, k)$ of MAX-E3-LIN-2 is NP-hard: given an instance of MAX-E3-LIN-2 with m equations and exactly k occuurrences of each variable, to decide if at least $(1+\varepsilon)m$ equations can be satisfied or if at most $(\frac{1}{2} - \varepsilon)m$ equations can be satisfied by an optimal assignment.*

To prove some inapproximability results, it is more convenient for reductions if all equations have the same right hand side. The results obtained in such case can be enforced by flipping some occurrences of variables ($\tilde{x} = 1 - x$). With that idea in mind, we can define a subproblem of $Q(\varepsilon, k)$.

Definition 6. *For $b \in \{0, 1\}$, we define $Q_b(\varepsilon, k)$ as the following partial decision problem:*

- ***Instance:*** *An instance of MAX-E3-LIN-2 with m equations of the form $x_{i_1} + x_{i_2} + x_{i_3} = b$, each variable occurring exactly k times negated and k times unnegated.*

- ***Objective:*** *Decide if at least $(1 + \varepsilon)m$ or at most $(\frac{1}{2} - \varepsilon)m$ equations can be satisfied by an optimal assignment.*

To continue with the reduction, we will extend the NP-hardness results to a low occurrence version of w-MAX-E3-LIN-2, a weighted system of linear equations over $\mathbb{Z}_2$ with either 2 or 3 variables. We call this a hybrid system of linear equations. Similarly, the task of the w-MAX-E3-LIN-2 problem is to find an assignment that maximizes the total weight of satisfied equations.

3 Reductions to w-MAX-E3-LIN-2

3.1 From $Q(\varepsilon, k)$ to w-MAX-E3-LIN-2

Let $\varepsilon \in (0, \frac{1}{4})$ and $k > 0$ be an integer such that the problem $Q(\varepsilon, k)$ is NP-hard. Let an instance I of $Q(\varepsilon, k)$ be given by a set $\nu(I)$ of variables, where $\nu := |\nu(I)|$. let's assume that there is an amplifier $G = (V, E)$ with edge weights $p : E \leftrightarrow \mathbb{R}^+$, with a set of contacts D with $|D| = k$.

Now, there is a gap preserving reduction from $Q(\varepsilon, k)$ to w-MAX-E3-LIN-2 with the amplifier as a parameter. The instance I of $Q(\varepsilon, k)$ is transformed into a hybrid instance J of w-MAX-E3-LIN-2 as follows:

- For each variable $x \in \nu(I)$, we create a copy of the amplifier G, denoted G_x.
  - Inside G_x, the vertices correspond to variables in J and each edge (u, v) represents the equation $u + v = 0$ with weight $p(u, v)$.

- Contacts in G_x represent the k occurrences of the variable x in the equations of I. Each contact is associated with one occurrence of x in I.

- Every equation in I of the form $x_{i_1} + x_{i_2} + x_{i_3} = b$ also belongs to J with weight 1.

Observe that the reduction above from I to J is a gap preserving reduction. Indeed, there is a dependency between the number of equations satisfied in I and the total weight of satisfied equations in J.

- If at least $(1+\varepsilon)m$ equations can be satisfied in I, then there is an assignment that satisfies all the equations in the amplifiers and at least $(1-\varepsilon)m$ equations of weight 1, leading to a total weight of satisfied equations of at least $W_1 = k \cdot p(E) + (1-\varepsilon)m$.

- If at most $(\frac{1}{2}-\varepsilon)m$ equations can be satisfied in I, then any assignment can satisfy at most $\frac{1}{2}m - \varepsilon m$ equations of weight 1, leading to a total weight of satisfied equations of at most $W_2 = k \cdot p(E) + (\frac{1}{2}-\varepsilon)m$.

If we look at these problems as minimum unsatisfiability problems, where OPT' is the minimum weight of unsatisfied equations over all assignments, then OPT'(I)= OPT'(J). Any assignment to variables from $\nu(I)$ generate an assignment to variables of J naturally, as the value of $x \in \nu(I)$ will be assigned to all the variables in G_x. These are called standard. Then, OPT'$(J) \leq$ OPT'(I) because any assignment to I can be extended to a standard assignment to J with the same weight of unsatisfied equations.
To see that the optimum OPT'(J) is attained on standard assignments, we use the properties of the amplifier. Any assignment φ to the variables of J can be transformed into a standard assignment φ' without increasing the weight of unsatisfied equations. For this, we assign to all variables in G_x the value as it is assigned to th majority of contact vertices in G_x by φ. The fact that G is an amplifier guarantees (by definition) that the weight of unsatisfied equations in J does not increase in this transformation. After iterating this process for all variables, the result will be a standard assignment without increasing the weight of unsatisfied equations in J. Therefore, OPT'(J) = OPT'(I).

3.2 From $Q_b(\varepsilon, 2k)$ to w-MAX-E3-LIN-2

The previous reduction is only the starting point, and a demonstration of how amplifiers can be useful. However, for the main results in this article, it is more convenient to start from the problem $Q_b(\varepsilon, k)$, where all equations have the same right hand side. This reduction is a slight modification of the previous one. Let $\varepsilon \in (0, \frac{1}{4})$ and $k > 0$ be an integer such that the problem $Q_b(\varepsilon, k)$ is NP-hard. Assume that $G = (V, E)$ is an amplifier with edge weights $p : E \leftrightarrow \mathbb{R}^+$ and a set of contacts D with $|D| = 2k$. Let $\{V^u, V^n\}$ be a partition of V balanced in D, so $\|D \cap V^u\| = \|D \cap V^n\| = k$. Also, let G^u and G^n be the induced subgraphs of G on V^u and V^n, respectively. Let an instance I of $Q_b(\varepsilon, 2k)$ be given by a set $\nu(I)$ of variables, where $\nu := |\nu(I)|$.

- For each variable $x \in \nu(I)$, we create a copy of the amplifier G, denoted G_x.
  - Any edge (u, u') in G_x^u or $G_x^{u'}$ represents the equation $u + u' = 0$ with weight $p(u, u')$. We call this cycle equations.
  - Any edge between $v \in V_x^u$ and $v' \in V_x^n$ represents the equation $v + v' = 1$ with weight $p(v, v')$. We call this matching equations.

- The contact vertices of G_x^u (and of G_x^n) represent the k occurrences of unnegated (respectively, negated) variable x in the equations of I. Every equation $x+y+z=b$ from I also belongs to J with weight 1.

This way, we produce an instance J of w-MAX-E3-LIN-2 with the same gap as in the previous reduction. As before, any assignment to variables from $\nu(I)$ generate an assignment to variables of J naturally, as the value of $x \in \nu(I)$ will be assigned to all the variables in G_x^u, and the value $1-x$ to all the variables in G_x^n. Once again, this assignment is called standard. The previous arguments applied to the problem $Q(\varepsilon, 2k)$ can be translated to this new setting, by reversing all variable from V_x^n to theis opposite.

3.3 Applying bi-wheel amplifiers

When we consider the existence of bi-wheel amplifiers $(W_{k,\tau}, p)$ and add themto the framework we have just defined, we obtain the following theorem:

Theorem 2 (Hybrid-$(W_{k,\tau}, p)$ inapproximability)**.** *For every $\varepsilon \in (0, \frac{1}{4}), b \in 0, 1$ there exist instances of w-MAX-E3-LIN-2, which we call hybrid-$(W_{k,\tau}, p)$, with the following properties:*

- *Each variable of the system of equations hybrid-$(W_{k,\tau}, p)$ occurs exactly 3 times.*
- *m equations are of the form $x+y+z=b$ with weight 1.*
- *$3\tau m$ equations are of the form $x+y=0$, with weight p_c.*
- *$\frac{3}{2}(\tau-1)m$ equations are of the form $x+y=1$, with weight p_m.*

It is NP-hard to decide if equations of weight at least $(1-\varepsilon)m$ or at most $(\frac{1}{2}-\varepsilon)m$ can be satisfied by an optimal assignment.

Proof. Let $\varepsilon \in (0, \frac{1}{4}), b \in 0, 1$ and $k > 0$ be an integer such that the problem $Q_b(\varepsilon, 2k)$ is NP-hard. Let an instance I of $Q_b(\varepsilon, 2k)$ be given by a set $\nu(I)$ of variables, where $\nu := |\nu(I)|$. Then $m := |I| = \frac{2k\nu}{3}$, as each variable occurs exactly $2k$ times and each equation has 3 variables. For any ficed variable $x \in \nu(I)$, we create a copy of the bi-wheel amplifier $W_{k,\tau}$, denoted W_x. Each of these is a copy of $W_{k,\tau}$, with $2\tau k$ variables $Var(x) = \{x_j^u, x_-j^n\}_{j=1}^{\tau k}$. Following the definition of a biwheel amplifier, each j^u is mapped to the variable x_j^u and j^n is mapped to x_j^n. We will denote the set of cycle edges as $\mathcal{C}^u(W_x)$ and $\mathcal{C}^n(W_x)$, and the set of matching edges as $\mathcal{M}(W_x)$.

- For each matching edge (j^u, j'^n) in $\mathcal{M}(W_x)$, we create an equation $x_j^u + x_{j'}^n = 1$ with weight p_m. There are $\frac{3}{2}(\tau-1)m$ such equations.
- For each cycle edge $(j^q, (j+1)^q)$ in $\mathcal{C}^q(W_x)$, we create an equation $x_j^q + x_{j+1}^q = 0$ with weight p_c. We have $2k\tau\nu = 3\tau m$ such equations.

Finally, we replace the l-th unnegated appearance of x in I with the variable $x_{c^u(l)}^u := c_l^u$, and the l-th negated appearance of x in I with the variable $x_{c^n(l)}^n := c_l^n$, for each $l \in \{1, \ldots, k\}$. This doesn't add equations to the system, but it just renames the variables in the equations of I to the variables in the bi-wheel amplifiers.

Then, we have produced m equations of the form $x+y+z=b$ with weight 1, $3\tau m$ equations of the form $x+y=0$, with weight p_c, and $\frac{3}{2}(\tau-1)m$ equations of the form $x+y=1$, with weight p_m. The degree of the variables is 3 in any case, as each variable in the bi-wheel amplifier has degree 3, and each variable in the equations of I is replaced by a variable in the bi-wheel amplifier that also has degree 3. The gap preserving property of the reduction is a consequence of the properties of the bi-wheel amplifier, as mentioned in the previous section. ☐

It is at this point that we change the perspective, as this is the problem we will be using from now onwards. We will start by labeling all the variables in the system of equations hybrid-$(W_{k,\tau}, p)$. First of all, there are $2\tau k\nu$ variables in the system, as each variable in $\nu(I)$ is replaced by $2\tau k$ variables in the bi-wheel amplifier. We label these variables as $x_{i,j,b}$, where $i \in \{1, \ldots, \nu\}$, $j \in \{1, \ldots, \tau k\}$ and $b \in \{u, n\}$. Each variable will participate in two distinct equations of the form $x + y = 0$, which correspond to the cycle equations. Additionally, each checker will appear in a single equation of the form $x + y = 1$ and every contact will appear in a single equation of the form $x + y + z = b$. Another way of expressing this notion is that for $S = \{\tau l : l \in \{1, \ldots, k\}\}$, and contacts are caracterized by $j \in S$, while checkers are caracterized by $j \notin S$. The weights of every equation are kept as in the definition of hybrid-$(W_{k,\tau}, p)$: p_c for cycle equations, p_m for matching equations and 1 for the equations of the form $x + y + z = b$.

In the second paper, they provide a different problem definition of w-MAX-E3-LIN-2. This definition has doubled the number of total clauses, and works with nk instead of m. This problem definition is equivalent, but the perspective on the inapproximability guarantee is slightly different. Setting $\varepsilon' = \frac{2}{3}\varepsilon$ and doubling the total amount of clauses:

Corollary 1. *It is NP-hard to decide if equations of weight at least $(\frac{2}{3} - \varepsilon')nk$ or at most $(1 - \varepsilon')nk$ can be satisfied by an optimal assignment.*

4 Reduction to TSP$_f$ from Hybrid-$(W_{k,\tau}, p)$

This reduction is defined by gadgets that determine how we map every equation to a graph in which we can define a valid tour. We call these **equation gadgets**. We have seen that there are three types of equations in our system. The most interesting ones are the equations of the form $x + y + z = b$, and they define degree 3 equation gadgets.

Definition 7 (Degree 3 equation gadget)**.** *An **degree 3 equation gadget** is a weighted multigraph $H = (V, E_u \dot\cup E_f, c)$ on a vertex set $V = [4 + n_{aux}]$ with the following properties:*

- *The vertices are partitioned into three sets:*
  - *A central vertex,* 4.
  - *A set of three contact vertices,* $\{1, 2, 3\}$.
  - *A set of auxiliary vertices,* $\{5, 6, \ldots, 4 + n_{aux}\}$.
- *H forms an instance of TSP$_f$, and contains a fixed set of unforced edges $E_s \subset E_u$, called special edges, of cost $1/2$ connecting the central vertex to the contacts.*

Given an equation gadget H and a valid tour Q in H, we define $S_i^H(Q)$ as the number of special edges in Q that appear i times in Q. We aim to encode an assignment $z \in \{0,1\}^3$ to the variables in an equation as a spanning tour such that the special edge connecting the central vertex to the i-th contact appears twice if $z_i = 1$ and does not appear if $z_i = 0$. The key idea is that satisfying assignments to the equation will correspond to valid tours with low weights and spanning tours not of this form will be more expensive. To encapsulate this, we define the following:

Definition 8 (Soundness and completeness)**.** *Given an equation gadget $H = (V, E_u \dot\cup E_f, w)$ for an equation $I = x + y + z = 1$, we define:*

- ***Completeness:*** *Let $z \in \{0,1\}^3$, and let $C \subset \{0,1\}^3$ be the set of satisfying assignments of I. We use $\mathcal{Q}_{\mathbf{z}}$ to refer to the set of valid tours Q in H such that the special edge $(\ell, 4)$ is unused for all ℓ such that $\mathbf{z}_\ell = 0$, and is used twice otherwise. $c(H)$ is defined as*

$$c(H) = \max_{z \in C} \min_{Q \in \mathcal{Q}_z} c(Q)$$

- ***Soundness:*** *$s(H)$ is defined as*

$$s(H) = \min_{Q} w(Q) - k_1^H(Q)/2 - \mathbb{1}_{k_1^H(Q)=0} \cdot \mathbb{1}_{k_2^H(Q) \text{ is even}}$$

where the minimum is taken over all valid spanning tours Q.

Now, we can provide a reduction from the NP-hard problem hybrid-$(W_{k,\tau}, p)$ to TSP_f. Defining $\Theta = \frac{1}{\max 1 p_m}$. Suppose we are given a Hybrid-$(W_{k,\tau}, p)$ instance I, we will create an instance G of TSP_f as follows:

- We will create a single central vertex s, as well as a vertex $v_{i,j,b}$ for each corresponding variable $x_{i,j,b}$ in the system of equations.
- For each equation of the form $x_{i,j,b} + x_{i',j',b'} = 0$, we add to G a single unforced edge of cost 1 between $v_{i,j,b}$ and $v_{i',j',b'}$.
- For each equation of the form $x_{i,j,b} + x_{i',j',b'} = 1$, we add to G two parallel forced edges of cost $2p_c\Theta$ between $v_{i,j,b}$ and $v_{i',j',b'}$.
- For each equation of the form $x_{i_1,j_1,b_1} + x_{i_2,j_2,b_2} + x_{i_3,j_3,b_3} = 1$, we add to G a copy of the gadget H corresponding to this equation, and we identify the contact vertices of H with the vertices v_{i_1,j_1,b_1}, v_{i_2,j_2,b_2} and v_{i_3,j_3,b_3}, and the central vertex of H with s. The central vertex is shared between all such gadgets.

Theorem 3 (Completeness)**.** *Suppose there is an assignment $x = (x_{i,j,b})_{i,j,b}$ to the Hybrid-$(W_{k,\tau}, p)$ instance that satisfies all degree 2 clauses and violates at most Δ degree 3 clauses. Then there is a spanning tour in G of total cost at most $nk(4(\tau-1)p_c\Theta + \tau + \frac{2}{3}c(H) - 1) + 2(\Delta + 2n)w(H)$.*

Proof. We construct collections of edges $T^{=}, T^{\neq}$ AND T^3 from G as follows:

- We add one copy of each forced edge of cost p_m to $T^{\neq}$.
- If a variable $x_{i,j,b}$ is set to 1, $T^{=}$ contains one copy of both unforced cycle edges of cost p_c adjacent to $v_{i,j,b}$. As the wheel contains two types of contact variables (negated and unnegated), one of the two sides of the wheel will be always connected.

The total cost of $T^{\neq}$ is $4 \cdot (\tau - 1)p_c\Theta nk$, while the total cost of $T^{=}$ is τnk. For each degree 3 clause $x_{i_1,j_1,b_1} + x_{i_2,j_2,b_2} + x_{i_3,j_3,b_3} = 1$ which is satisfied, we consider an optimal valid tour $Q* = \arg\min_{Q \in Q_z} c(Q)$. Then, we add the non-special edges of Q^* to T^3. As an exception, in the case that j_1, j_2, or $j_3 = 0$ we will add all special edges, to ensure connectivity. This has a total cost of 3 per wheel, so $3n$ in total. It will be negligible for high values of k. This will ensure connectivity. If it is not satisfied, we add all a minimum cost tour, as well as all the special edges twice. The cost of edges added per degree 3 gadget is then bounded by:

$$c(H) - (x_{i_1,j_1,b_1} + x_{i_2,j_2,b_2} + x_{i_3,j_3,b_3}) + 2w(H)(\mathbb{1}_{0 \in \{j_1,j_2,j_3\}} + \mathbb{1}_{x_{i_1,j_1,b_1} + x_{i_2,j_2,b_2} + x_{i_3,j_3,b_3} = 0})$$

As an explanation of the last term, if the clause is satisfied, we add at most $c(H)$, while if it is not satisfied, we add at most $c(H) + 2w(H)$, as we will need to add all special edges to ensure connectivity. Summing over all degree three clauses, we obtain $w(T^3) \leq c(H) - nk + 2w(H)(2n + \delta)$. The sum over all assignments is exactly nk because all the contact variables appear exactly one time per equation, half of them have value 0 and half of them have value 1 and there's $2nk$ of them. The last term references the total number of unsatisfied clauses (Δ) and the fact that at most $2n$ clauses contain the first element of a wheel. This subset of edges that we have just built keeps an even degree over all vertices, as the three ways in which we have added vertices do.

Claim 1. *T has a single connected component consisting of $V(G)$*

Proof. We will argue that all the wheels are connected to the central vertex s. In the biwheel structure, all the checkers are connected to one checker in the other wheel via two forced edges. Also, if a variable is set to a given value $b \in 0, 1$, the corresponding wheel is connected via the unforced edges, connecting all the checkers and half the contacts. These are connected to the central vertex via the special degree 3 tour in the first vertex per wheel. Additionally, all the satisfied equations have at least one variable set to 1, so all the contacts assigned 0 that appear in satisfied equations are connected, and the unsatisfied equations are connected. What remains to be seen is that the auxiliary vertices are connected.

The minimum cost tour over all the degree 3 gadgets covers all of them, so trivially this condition is also fulfilled. □

By construction, all the forced edges are used at least once and none is used more than twice. As it is connecting and spanning, it is a valid tour. Its total cost is

$$c(T^{\neq}) + c(T^{=}) + c(T^3) \leq 4(\tau - 1)p_c\Theta nk + \tau nk + c(H) - nk + 2w(H)(2n + \delta) =$$

$$= nk(4(\tau - 1)p_c\Theta + \tau + \frac{2}{3}c(H) - 1) + 2(\Delta + 2n)w(H)$$

□

Theorem 4 (Soundness)**.** *Suppose there is a valid spanning tour T of the TSP_f instance G with cost at most $nk(2/3s(H) + (2p_c\Theta + 1)2(\tau - 1)) + \Delta$. Then, there is an assignment that leaves at most Δ clauses unsatisfied.*

Proof. Let us, as before, write $T = T^{\neq} + T^{=} + T^3$, where each of the components is, as before, the amount of edges in T belonging to every part of the reduction. We define $U^{=}$ to be $T^{=}$ without the edges appearing twice removed. This forms a quasi-tour. To assign values to variables, we will look at its local value in $U^{=}$. In particular, for a given variable $x_{i,j,b}$ we denote $l_{i,j,b}$ and $r_{i,j,b}$ to be the number of copies of $(x_{i,j-1,b}, x_{i,j,b})$ and $(x_{i,j,b}, x_{i,j+1,b})$, respectively. Of course, $l_{i,j,b}, r_{i,j,b} \in \{0, 1\}$. We say $x_{i,j,b}$ is *honest* if the $l_{i,j,b} = r_{i,j,b}$, and *dishonest* otherwise. It is quite easy to define the value of a honest variable. We will assign $x_{i,j,b} = l_{i,j,b}$. However, when assigning values to a dishonest variable, one must be a bit more careful.

Claim 2. *Two edges connected by an inequality clause are either both honest or both dishonest*

Proof. The only edges adjacent to a pair of vertices $x_{i,j,u}, x_{i,j,n}$ are the two forced edges connecting them and the two cycle edges connecting each of them to their respective cycle. As the forced edges are adjacent to both of the vertices, removing them doesn't change parity of the degree difference. Let's assume, w.l.o.g., that $x_{i,j,u}$ is honest and $x_{i,j,n}$ is dishonest. Then, $\deg(x_{i,j,u}) - \deg(x_{i,j,n})$ is odd, which means one of the two has odd degree originally. This contradicts T being a tour. □

- If both edges connected to an inequality clause are dishonest, we assign a random value $b \in 0, 1$ to $x_{i,j,u}$ and the opposite value to $x_{i,j,n}$.
- For each degree 3 clause containing at least one dishonest variable, we assign a value to said variable that ensures the clause is satisfied.

Once we have defined the value each variable takes, we define some quantities:

- d_{ch}: number of dishonest checkers

- d_{co}: number of dishonest contacts
- p_{ch}: number of inequality constraints with honest contacts and same value assigned
- r: number of unsatisfied degree 3 equation with honest variables

Claim 3. *The expected weight of unsatisfied clauses is at most* $p_c(d_{ch} + d_{co}) + p_m p_{ch} + r$

Proof. Let's upper bound the total number of clauses by recapping our previous analysis.

- Equality clauses are always satisfied in honest vertices. Every equality clause adjacent to a dishonest checker is satisfied with probability 1/2 due to the random assignment. That also takes care of the equality constraints of dishonest checkers adjacent to contacts. If the adjacent checkers are honest, they must be assigned different values, as their contact is dishonest. Then, only one is unsatisfied. Then, the expected number of equality clauses is at most $d_{ch} + d_{co}$.
- Inequality clauses are only unsatisfied when the vertices they contain are both honest and have the same value. These are just p_{ch}.
- All clauses with dishonest vertices are satisfied, so the total number of unsatisfied clauses is r.

After taking into account the weights of each equation, the result is proven. □

Claim 4. $w(T) \geq nk \cdot (30 + 2 \cdot s(H)/3) + (d_{ch} + d_{co})/2 + p_{ch} + r$

Proof. The idea is separately bounding the cost of edges in $T^=$, $T^{\neq}$ and T^3 separately. For this, once again, we will need to define a few quantities.

- t_{co}: number of honest contacts assigned 1.
- p^b_{ch}: number of inequality clauses in which both checkers are assigned $b \in \{0, 1\}$. Of course, $p^0_{ch} + p^1_{ch} = p_{ch}$.
- u_{co}: number of connected components of $U^= \cup T^{\neq} \cup T^3$ using only edges from T^3

Then, we can commence the bounding:

- $U^=$: We can write

$$|U^=| = \frac{1}{2}\sum_{i,j,b}(l_{i,j,b} + r_{i,j,b}) = \frac{1}{2}\left(\sum_{i,j:j|\tau,b}(l_{i,j,b} + r_{i,j,b}) + \sum_{i,j:j\nmid\tau} v_{i,j}\right)$$

  This is just a separation of the sum, using $v_{i,j} = l_{i,j,u} + r_{i,j,u} + l_{i,M(j),n} + r_{i,M(j),n}$. The left part of the sum affects the contact variables and the right one the checker variables. To bound the sum over all contacts, we note that, as we have proven before, a dishonest contact variable has $l_{i,j,b} + r_{i,j,b}$=1 and a honest variable assigned 1 has $l_{i,j,b} + r_{i,j,b}$=2. Then the first sum is $d_{co} + 2t_{co}$. To bound the second sum, we see that, as per claim2, $v_{i,j}$ must be even. Then $v_i, j = 2$ if both checkers are dishonest, $v_i, j = 0$ if both checkers are honest and assigned 0 and $v_{i,j} = 4$ if they are honest and assigned 1. Therefore, the left part of the sum is $2(\tau - 1)nk + 2p^1_{ch} - 2p^0_{ch}$. Using the fact that each edge was cost p_c we obtain

$$c(U^=) = ((\tau - 1)nk + p^1_{ch} - p^0_{ch} + \frac{d_{co}}{2} + t_{co})$$

- $T^{=}\backslash U^{=}$: For this, since T is a valid spanning tour, the total cost of edges in $T^{=}\backslash U^{=}$ is at least the number of connected components in $U^{=} \cup T^{\neq} \cup T^3$ (two edges connecting them, each of them of cost p_c).
$$c(T^{=}\backslash U^{=}) \geq 2(p_{ch}^0 + u_{co})$$
- $T^{\neq}$: For any given pair of checker vertices connected, the tour crosses each forced edge at least once. Also, if either of the vertices is dishonest, T being a tour implies one is crossed an additional time. Thus, $w(T^{\neq}) \geq 2p_c\Theta(2(\tau-1)nk + d_{ch}/2)$.
- T^3: For a given degree 3 clause C involving contacts $(x_{i_1,j_1,b_1}, x_{i_2,j_2,b_2}, x_{i_3,j_3,b_3})$, we can denote T_C as the edges in T^3 that appear in the gadget corresponding to that clause. The main idea is extracting a valid tour Q of the gadget graph H and then use soundness properties of that graph to argue that $w(T_C)$ is large. First, we will take all the edges from T_C into Q. Still, Q might not be a spanning tour. To solve this, we add $l_{i_l,j_l,b_l} + r_{i_l,j_l,b_l}$ copies of the speciel edge $(l, 4)$ for edge $l \in \{1, 2, 3\}$, to ensure even degree and increase connectivity. For eah connected component in Q not connected to the central vertex, we will add two copies of the special edge $l, 4$ for each such component. We will denote u_{co}^{local} the number of such components. Observe that the degree of all auxiliary vertices in T_C is the same as in Q, and they are even. Since we have made sure tu preserve even degree when adding edges, all vertices have even degree. Also, we have ensured connectivity. As Q uses each edge at most twice, forced edges are used at least once and is a tour, it is a valid tour. Then, by definition of soundness:
$$c(Q) \geq s(H) + k_1^H(Q)/2 + \mathbb{1}_{k_1^H(Q)=0} \cdot \mathbb{1}_{k_2^H(Q) \text{ is even}}$$
Subtracting the contribution of the edges we added to Q, we obtain:
$$c(T_C) = c(Q) - k_1^H(Q) - k_2^H(Q) \geq -u \geq s(H) - k_2^H(Q) + \mathbb{1}_{k_1^H(Q)=0} \cdot \mathbb{1}_{k_2^H(Q) \text{ is even}}$$
Note that if $u_{co}^{local} = 0$, the indicator exactly indicates whether the clause is satisfied. Then $s(H) - k_2^H(Q) + \mathbb{1}_{k_1^H(Q)=0} \cdot \mathbb{1}_{k_2^H(Q) \text{ is even}} \geq s(H) - u_{co}^{local} - k_2^H + C$. Here, $C = 0$ if the clause isn't satisfied (following the two special edges if 1, 0 otherwise criterion). Summing over all $2nk/3$ degree 3 clauses and using the fact that the sum of $k_2^H - u_{co}^{local}$ is the total number of honest contact vertices assigned 1, t_{co}.
$$c(T^3) \geq 2nk/3s(H) - 2u_{co} - t_{co} + r$$

Then the lower bound is
$$c(T) = c(T^{=}) + c(T^{\neq}) + c(T^3) \geq nk(2/3s(H) + (2p_c\Theta + 1)2(\tau - 1)) + \frac{d_{co} + 2p_c\Theta d_{ch}}{2} + r + p_{ch}$$
□

Supposing $p_m \leq 1$ (that is equivalent to $p_m = 1$), then as $p_c \geq 1/2$, $\frac{d_{co}+2p_c\Theta d_{ch}}{2} + r + p_{ch} \leq p_c * (d_{co} + d_{ch}) + r + p_{ch} = \Delta$. The assignment we have built this way has the properties we desire. □

Theorem 5. *Let H be an equation gadget such that $c(H) < \infty$. For any $\varepsilon > 0$, it is NP-hard to approximate TSP_f within x*

Proof. Using theorems 2, 3 and 4, we deduce that it is NP-hard to distinguish between instances that admit a spanning tour of weight at most $nk(4(\tau-1)p_c\Theta+\tau+\frac{2}{3}c(H)-1)+2(2/3nk\varepsilon+2n)w(H)$ from instances where all spanning tours have weight at least □